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MANUFACTURING METHOD OF SEMICONDUCTOR STRUCTURE INCLUDING SEAL RING STRUCTURE

Abstract

A manufacturing method of a semiconductor structure includes following steps. A seal ring structure is formed. A method of forming the seal ring structure includes following steps. A first interconnect structure is formed in a first dielectric layer. A second dielectric layer is formed on the first interconnect structure and the first dielectric layer. A first opening exposing the first interconnect structure and a first ditch exposing the first dielectric layer are formed in the second dielectric layer. A second interconnect structure is formed in the first opening, and the second interconnect structure is connected to the first interconnect structure. A passivation layer is formed on the seal ring structure and the second dielectric layer and in the first ditch. A second ditch is formed. The second ditch passes through the passivation layer located in the first ditch and extends into the first dielectric layer below the first ditch.

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Background/Summary

CROSS-REFERENCE TO RELATED APPLICATION [0001] This application is a divisional application of and claims the priority benefit of U.S. application Ser. No. 17/735,099, filed on May 2, 2022, which claims the priority benefit of China patent application serial no. 202210375520.5, filed on Apr. 11, 2022. The entirety of each of the above-mentioned patent applications is hereby incorporated by reference herein and made a part of this specification.

BACKGROUND

Technical Field

[0002] The disclosure relates to a semiconductor structure and a manufacturing method thereof; more particularly, the disclosure relates to a semiconductor structure with a ditch and a manufacturing method thereof.

Description of Related Art

[0003] The existing semiconductor structure may produce metal burr when a laser grooving process is being performed, and the metal burr may lead to unnecessary short circuit during a packaging process, thus resulting in a decrease in the product yield.

[0004] The solution to the above issue at present is to form a ditch in the semiconductor structure, whereby the metal burr falls into the ditch to prevent the unnecessary short circuit caused by the metal burr during the packaging process. Since the ditch has a relatively large depth, an etching process for forming the ditch requires a relatively long period of time. To shorten the time of the etching process for forming the ditch, the power of the etching process should be increased. However, the high-power etching process may bring about a negative impact on electrical properties of semiconductor devices.

SUMMARY

[0005] The disclosure provides a semiconductor structure and a manufacturing method thereof which may reduce time and power of an etching process for forming a ditch.

[0006] In an embodiment of the disclosure, a semiconductor structure includes a substrate, a first dielectric layer, a second dielectric layer, a seal ring structure, and a passivation layer. The first dielectric layer is located on the substrate. The second dielectric layer is located on the first dielectric layer. The seal ring structure includes a first interconnect structure and a second interconnect structure. The first interconnect structure is located in the first dielectric layer. The second interconnect structure is located in the second dielectric layer and connected to the first interconnect structure. The passivation layer is located on the seal ring structure and the second dielectric layer. The passivation layer has a spacer portion. The spacer portion covers a sidewall of the second dielectric layer and a portion of the first dielectric layer. A ditch exists in the passivation layer and the first dielectric layer. The spacer portion is located between the ditch and the seal ring structure.

[0007] According to an embodiment of the disclosure, the substrate may include a seal ring region and a scribe line region which are adjacent to each other.

[0008] According to an embodiment of the disclosure, the seal ring structure may be located in the

seal ring region, and the ditch may be located in the scribe line region.

[0009] According to an embodiment of the disclosure, the ditch may expose a sidewall of the spacer portion.

[0010] According to an embodiment of the disclosure, a bottom surface of the ditch may be lower than a bottom surface of the spacer portion.

[0011] According to an embodiment of the disclosure, a bottom surface of the ditch may be lower than a bottom surface of the second interconnect structure.

[0012] According to an embodiment of the disclosure, the passivation layer may have an opening therein, and the opening may expose the second interconnect structure.

[0013] According to an embodiment of the disclosure, the second interconnect structure may be further located on a top surface of the second dielectric layer.

[0014] According to an embodiment of the disclosure, the passivation layer may have a single-layer structure.

[0015] According to an embodiment of the disclosure, the passivation layer may have a multi-layer structure.

[0016] In an embodiment of the disclosure, a manufacturing method of a semiconductor structure includes following steps. A substrate is provided. A first dielectric layer is formed on the substrate. A seal ring structure is formed. A method of forming the seal ring structure includes following steps. A first interconnect structure is formed in the first dielectric layer. A second dielectric layer is formed on the first interconnect structure and the first dielectric layer. A first opening and a first ditch are formed in the second dielectric layer. The first opening exposes the first interconnect structure, and the first ditch exposes the first dielectric layer. A second interconnect structure is formed in the first opening, and the second interconnect structure is connected to the first interconnect structure. A passivation layer is formed on the seal ring structure and the second dielectric layer and in the first ditch. A second ditch is formed. The second ditch passes through the passivation layer located in the first ditch and extends into the first dielectric layer below the first ditch.

[0017] According to another embodiment of the disclosure, the substrate may include a seal ring region and a scribe line region which are adjacent to each other.

[0018] According to another embodiment of the disclosure, the seal ring structure may be located in the seal ring region, and the second ditch may be located in the scribe line region.

[0019] According to another embodiment of the disclosure, the passivation layer may have a spacer portion, and the spacer portion may cover a sidewall of the second dielectric layer and a portion of the first dielectric layer.

[0020] According to another embodiment of the disclosure, the spacer portion may be located between the second ditch and the seal ring structure.

[0021] According to another embodiment of the disclosure, a width of the second ditch may be smaller than a width of the first ditch.

[0022] According to another embodiment of the disclosure, the first opening and the first ditch may be formed simultaneously.

[0023] According to another embodiment of the disclosure, a portion of the first dielectric layer may be removed while the second interconnect structure is being formed.

[0024] According to another embodiment of the disclosure, a bottom surface of the second ditch may be lower than a bottom surface of the second interconnect structure.

[0025] According to another embodiment of the disclosure, the manufacturing method of the semiconductor structure may further include following steps. A second opening is formed in the passivation layer. The second opening may expose the second interconnect structure, and the second opening and the second ditch may be formed simultaneously.

[0026] In view of the above, in the semiconductor structure provided in one or more embodiments of the disclosure, the spacer portion of the passivation layer covers the sidewall of the second

dielectric layer and a portion of the first dielectric layer. That is, before the ditch is formed, a portion of the second dielectric layer may be removed, whereby the time and the power of the etching process for forming the ditch may be reduced. Besides, owing to the possibility of reducing the power of the etching process for forming the ditch, a negative impact of the etching process on semiconductor devices may be prevented.

[0027] Additionally, in the manufacturing method of the semiconductor structure provided in one or more embodiments of the disclosure, the first ditch is formed in the second dielectric layer, and then the second ditch is formed. Namely, before the second ditch is formed, a portion of the second dielectric layer is already removed, whereby the time and the power of the etching process for forming the second ditch may be reduced. Besides, owing to the possibility of reducing the power of the etching process for forming the second ditch, a negative impact of the etching process on semiconductor devices may be prevented.

[0028] To make the above more comprehensible, several embodiments accompanied with drawings are described in detail as follows.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

[0029] The accompanying drawings are included to provide a further understanding of the disclosure, and are incorporated in and constitute a part of this specification. The drawings illustrate embodiments of the disclosure and, together with the description, serve to explain the principles of the disclosure.

[0030] FIG. 1A to FIG. 1D are cross-sectional views of a manufacturing process of a semiconductor structure according to some embodiments of the disclosure.

[0031] FIG. 2 is a cross-sectional view of a semiconductor structure according to other embodiments of the disclosure.

DESCRIPTION OF THE EMBODIMENTS

[0032] FIG. 1A to FIG. 1D are cross-sectional views of a manufacturing process of a semiconductor structure according to some embodiments of the disclosure. FIG. 2 is a cross-sectional view of a semiconductor structure according to other embodiments of the disclosure.

[0033] With reference to FIG. 1A, a substrate **100** is provided. The substrate **100** may include a seal ring region R1 and a scribe line region R2 that are adjacent to each other. The substrate **100** may be a semiconductor substrate, such as a silicon substrate. In addition, although not shown in FIG. 1A, the substrate **100** may have required components therein, such as doped regions and/or isolation structures, and the substrate **100** may have required components thereon, such as semiconductor devices (e.g., active devices and/or passive devices); the relevant descriptions of these components are omitted here.

[0034] A dielectric layer **102** is formed on the substrate **100**. The dielectric layer **102** may have a single-layer structure or a multi-layer structure. A material of the dielectric layer **102** is, for instance, silicon oxide, silicon nitride, or a combination thereof. A method of forming the dielectric layer **102** is, for instance, chemical vapor deposition (CVD).

[0035] With reference to FIG. 1A and FIG. 1B, a seal ring structure SR (shown in FIG. 1B) is formed. In some embodiments, the seal ring structure SR may be a die seal ring structure. The seal ring structure SR may be located in the seal ring region R1. A method of forming the seal ring structure SR includes following steps. With reference to FIG. 1A, an interconnect structure **104** is formed in the dielectric layer **102**. A material of the interconnect structure **104** is, for instance, metal, such as copper. The interconnect structure **104** may be formed by performing a metal interconnect process. The number of layers of the interconnect structure **104** is not limited to the number of layers shown in the drawings. Those having ordinary knowledge in the pertinent field

may adjust the number of layers of the interconnect structure **104** according to actual requirements. [0036] Next, a dielectric layer **106** is formed on the interconnect structure **104** and the dielectric layer **102**. The dielectric layer **106** may have a single-layer structure or a multi-layer structure. A material of the dielectric layer **106** is, for instance, silicon oxide, silicon nitride, or a combination thereof. A method of forming the dielectric layer **106** is, for instance, CVD.

[0037] An opening **OP1** and a ditch **D1** are formed in the dielectric layer **106**. The opening **OP1** exposes the interconnect structure **104**. The ditch **D1** exposes the dielectric layer **102**. The opening **OP1** may be located in the seal ring region **R1**. The ditch **D1** may be located in the scribe line region **R2**. A method of forming the opening **OP1** and the ditch **D1** is, for instance, patterning the dielectric layer **106** by performing a photolithography process and an etching process. In some embodiments, the opening **OP1** and the ditch **D1** may be formed simultaneously. In some embodiments, during the process of forming the opening **OP1** and the ditch **D1**, a portion of the dielectric layer **102** may be removed.

[0038] With reference to FIG. **1B**, an interconnect structure **108** is formed in the opening **OP1**. The interconnect structure **108** is connected to the interconnect structure **104**. In some embodiments, the interconnect structure **108** may be further located on a top surface **TS** of the dielectric layer **106**. In some embodiments, in a process of forming the interconnect structure **108**, a portion of the dielectric layer **102** may be removed, whereby a bottom surface **BS1** of the ditch **D1** may be lower than a bottom surface **BS2** of the interconnect structure **108**. In some embodiments, during the process of forming the interconnect structure **108**, a portion of the dielectric layer **106** may be removed. A material of the interconnect structure **108** is, for instance, aluminum and other metals. The interconnect structure **108** may be formed by applying a metal interconnect process. The interconnect structure **108** is not limited to the structure shown in the drawings. Those having ordinary knowledge in the pertinent field may adjust the interconnect structure **108** according to actual requirements.

[0039] Through applying the aforesaid method, the seal ring structure **SR** may be formed. The seal ring structure **SR** includes the interconnect structure **104** and the interconnect structure **108**. In this embodiment, the seal ring structure **SR** may include a seal ring **SR1** and a seal ring **SR2**. For instance, the seal ring **SR1** may include an interconnect structure **104a** and the interconnect structure **108**, and the seal ring **SR2** may include an interconnect structure **104b**, which should however not be construed as a limitation in the disclosure. In this embodiment, the seal ring structure **SR** is exemplified by including a plurality of seal rings (e.g., the seal ring **SR1** and the seal ring **SR2**), which should however not be construed as a limitation in the disclosure. As long as the seal ring structure **SR** includes at least one seal ring, it falls within the scope of protection provided in the disclosure.

[0040] With reference to FIG. **1C**, a passivation layer **110** is formed on the seal ring structure **SR** and the dielectric layer **106** and in the ditch **D1**. The passivation layer **110** may have a single-layer structure or a multi-layer structure. In this embodiment, the passivation layer **110** is exemplified by having the multi-layer structure, which should however not be construed as a limitation in the disclosure. For instance, the passivation layer **110** may include a passivation layer **110a** and a passivation layer **110b**. The passivation layer **110a** is located on the seal ring structure **SR** and the dielectric layer **106** and in the ditch **D1**. A material of the passivation layer **110a** is, for instance, silicon oxide. A method of forming the passivation layer **110a** is, for instance, CVD. The passivation layer **110b** is located on the passivation layer **110a**. A material of the passivation layer **110b** is, for instance, silicon nitride. A method of forming the passivation layer **110b** is, for instance, CVD.

[0041] With reference to FIG. **1D**, a ditch **D2** is formed. The ditch **D2** may be located in the scribe line region **R2**. A width **W2** of the ditch **D2** may be smaller than a width **W1** of the ditch **D1**. A bottom surface **BS3** of the ditch **D2** may be lower than a bottom surface **BS2** of the interconnect structure **108**. A method of forming the ditch **D2** is, for instance, patterning the passivation layer

110 and the dielectric layer **102** by performing a photolithography process and an etching process. In addition, an opening **OP2** may be further formed in the passivation layer **110**. The opening **OP2** may be located in the seal ring region **R1**. The opening **OP2** may expose the interconnect structure **108**. A method of forming the opening **OP2** is, for instance, patterning the passivation layer **110** by performing a photolithography process and an etching process. In some embodiments, the opening **OP2** and the ditch **D2** may be formed simultaneously.

[0042] The ditch **D2** passes through the passivation layer **110** located in the ditch **D1** and extends into the dielectric layer **102** below the ditch **D1**. The passivation layer **110** may have a spacer portion **SP**. The spacer portion **SP** may cover a sidewall **SW1** of the dielectric layer **106** and a portion of the dielectric layer **102**. The spacer portion **SP** may be located between the ditch **D2** and the seal ring structure **SR**. A bottom surface **BS4** of the spacer portion **SP** may be lower than the bottom surface **BS2** of the interconnect structure **108**. The ditch **D2** may expose a sidewall **SW2** of the spacer portion **SP**. The bottom surface **BS3** of the ditch **D2** may be lower than the bottom surface **BS4** of the spacer portion **SP**.

[0043] According to the previous embodiments, in the manufacturing method of the semiconductor structure, the ditch **D1** is formed in the dielectric layer **106**, and then the ditch **D2** is formed. Namely, before the ditch **D2** is formed, a portion of the dielectric layer **106** is already removed, whereby the time and the power of the etching process for forming the ditch **D2** may be reduced. Besides, owing to the possibility of reducing the power of the etching process for forming the ditch **D2**, a negative impact of the etching process on semiconductor devices may be prevented.

[0044] Hereinafter, the semiconductor structure **10** provided in the previous embodiment is described with reference to FIG. **1D**. Besides, although the method of forming the semiconductor structure **10** is exemplified by the above-mentioned method, the exemplified method should however not be construed as a limitation in the disclosure.

[0045] With reference to FIG. **1D**, the semiconductor structure **10** includes the substrate **100**, the dielectric layer **102**, the dielectric layer **106**, the seal ring structure **SR**, and the passivation layer **110**. The dielectric layer **102** is located on the substrate **100**. The dielectric layer **106** is located on the dielectric layer **102**. The seal ring structure **SR** includes the interconnect structure **104** and interconnect structure **108**. The interconnect structure **104** is located in the dielectric layer **102**. The interconnect structure **108** is located in the dielectric layer **106** and connected to the interconnect structure **104**. The passivation layer **110** is located on the seal ring structure **SR** and the dielectric layer **106**. The passivation layer **110** has the spacer portion **SP**. The spacer portion **SP** covers the sidewall **SW1** of the dielectric layer **106** and a portion of the dielectric layer **102**. The ditch **D2** exists in the passivation layer **110** and the dielectric layer **102**. The spacer portion **SP** is located between the ditch **D2** and the seal ring structure **SR**. In this embodiment, the passivation layer **110** may have the opening **OP2** which may expose the interconnect structure **108**, which should however not be construed as a limitation in the disclosure. In another embodiment, as shown in FIG. **2**, the opening **OP2** in FIG. **1** may not be included in the passivation layer **110** in FIG. **2**.

[0046] In addition, the detailed contents of each component in the semiconductor structure **10** (e.g., the material, the arrangement relationship, the forming method, the effects, etc.) have been elaborated in the previous embodiments and thus will not be repeated hereinafter.

[0047] As provided in the previous embodiments, in the semiconductor structure **10**, the spacer portion **SP** of the passivation layer **110** covers the sidewall **SW1** of the dielectric layer **106** and a portion of the dielectric layer **102**. That is, before the ditch **D2** is formed, a portion of the dielectric layer **106** may be removed, whereby the time and the power of the etching process for forming the ditch **D2** may be reduced. Besides, owing to the possibility of reducing the power of the etching process for forming the ditch **D2**, the negative impact of the etching process on the semiconductor devices may be prevented.

[0048] To sum up, in the semiconductor structure and the manufacturing method thereof provided in one or more embodiments of the disclosure, the time and the power of the etching process for

forming the ditch may be reduced. Besides, owing to the possibility of reducing the power of the etching process for forming the ditch, the negative impact of the etching process on the semiconductor devices may be prevented.

[0049] It will be apparent to those skilled in the art that various modifications and variations may be made to the disclosed embodiments without departing from the scope or spirit of the disclosure. In view of the foregoing, it is intended that the disclosure covers modifications and variations provided they fall within the scope of the following claims and their equivalents.

Claims

1. A manufacturing method of a semiconductor structure, comprising: providing a substrate; forming a first dielectric layer on the substrate; forming a seal ring structure, wherein a method of forming the seal ring structure comprises: forming a first interconnect structure in the first dielectric layer; forming a second dielectric layer on the first interconnect structure and the first dielectric layer; forming a first opening and a first ditch in the second dielectric layer, wherein the first opening exposes the first interconnect structure, and the first ditch exposes the first dielectric layer; and forming a second interconnect structure in the first opening, wherein the second interconnect structure is connected to the first interconnect structure; and forming a passivation layer on the seal ring structure and the second dielectric layer and in the first ditch; and forming a second ditch, wherein the second ditch passes through the passivation layer located in the first ditch and extends into the first dielectric layer below the first ditch.
 2. The manufacturing method according to claim 1, wherein the substrate comprises a seal ring region and a scribe line region adjacent to each other.
 3. The manufacturing method according to claim 2, wherein the seal ring structure is located in the seal ring region, and the second ditch is located in the scribe line region.
 4. The manufacturing method according to claim 1, wherein the passivation layer has a spacer portion, and the spacer portion covers a sidewall of the second dielectric layer and a portion of the first dielectric layer.
 5. The manufacturing method according to claim 4, wherein the spacer portion is located between the second ditch and the seal ring structure.
 6. The manufacturing method according to claim 1, wherein a width of the second ditch is smaller than a width of the first ditch.
 7. The manufacturing method according to claim 1, wherein the first opening and the first ditch are formed simultaneously.
 8. The manufacturing method according to claim 1, wherein a portion of the first dielectric layer is removed while the second interconnect structure is being formed.
 9. The manufacturing method according to claim 1, wherein a bottom surface of the second ditch is lower than a bottom surface of the second interconnect structure.
 10. The manufacturing method according to claim 1, further comprising: forming a second opening in the passivation layer, wherein the second opening exposes the second interconnect structure, and the second opening and the second ditch are formed simultaneously.
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